0.1 Quantum computation as controlled dynamics

In the abstract circuit model, a quantum algorithm is described as a sequence of unitary gates acting on an n-qubit register. From the hardware point of view, however, these gates are not primitive objects: they arise from the real-time evolution of a physical quantum system driven by externally programmable fields. Quantum control is therefore the discipline that bridges the algorithmic layer of quantum information processing and the dynamical laws governing a concrete device. In this perspective, the implementation of a gate is naturally formulated as an inverse problem: given a desired logical operation, one seeks a family of control signals that steers the system so that its physical evolution reproduces the target operation with high accuracy on the computational degrees of freedom [**?**, **?**, **?**, **?**].

For an n-qubit processor, the logical information is encoded in the computational subspace

$$\mathcal{H}_{\text{comp}} \cong (\mathbb{C}^2)^{\otimes n} \subseteq \mathcal{H}_{\text{phys}}, \tag{1}$$

where $\mathcal{H}_{\text{phys}}$ denotes the full Hilbert space of the device. The inclusion in Eq. (1) is essential: in realistic platforms, the physical Hilbert space is generally larger than the ideal qubit space. This point is conceptually important already at the beginning of the thesis, because it clarifies that a quantum gate is not merely a map on $\mathcal{H}_{\text{comp}}$, but the effective restriction of a higher-dimensional physical evolution to the logical sector. In superconducting architectures, for instance, this enlarged Hilbert space is associated with higher oscillator levels beyond the two-level qubit approximation, an observation that will later motivate the discussion of leakage errors [**?**, **?**].

A broad and standard model for controlled quantum dynamics is a time-dependent Hamiltonian of the form

$$H(t;\mathbf{u}) = H_{\text{d}} + \sum_{k=1}^{m} u_k(t)\, H_k, \tag{2}$$

where H_{d} is the drift Hamiltonian, fixed by the underlying hardware, while the operators $\{H_k\}_{k=1}^{m}$ specify the controllable interaction channels and the real functions

$$\mathbf{u}(t) = \big(u_1(t), u_2(t), \ldots, u_m(t)\big) \tag{3}$$

are the externally applied control fields. Depending on the platform, the components of $\mathbf{u}(t)$ may represent microwave amplitudes, phases, detunings, flux biases, or other tunable experimental knobs. The state vector then obeys the time-dependent Schrödinger equation

$$i\hbar \frac{d}{dt} \left|\psi(t)\right\rangle = H(t;\mathbf{u}) \left|\psi(t)\right\rangle, \tag{4}$$

or, equivalently, the propagator $U(t,0)$ satisfies

$$i\hbar \frac{d}{dt} U(t,0) = H(t;\mathbf{u})\, U(t,0), \qquad U(0,0) = \mathbb{I}, \tag{5}$$

with formal solution

$$U(T,0) = \mathcal{T} \exp\left[-\frac{i}{\hbar} \int_0^T H(t;\mathbf{u})\; dt \right], \tag{6}$$

where $\mathcal{T}$ denotes the time-ordering operator [**?**, **?**, **?**].

The physical task of gate synthesis can now be stated precisely. Let Π_{comp} be the projector onto $\mathcal{H}_{\text{comp}}$, and let $U_{\text{target}} \in U(2^n)$ be the desired logical gate. The effective action induced on the computational sector is

$$U_{\text{eff}}(T) := \Pi_{\text{comp}}\, U(T,0)\, \Pi_{\text{comp}}, \tag{7}$$

and the control problem consists in selecting the fields $\mathbf{u}(t)$ and, in many cases, the gate duration T, so that

$$U_{\text{eff}}(T) \approx e^{i\phi} U_{\text{target}}, \qquad \phi \in \mathbb{R}, \tag{8}$$

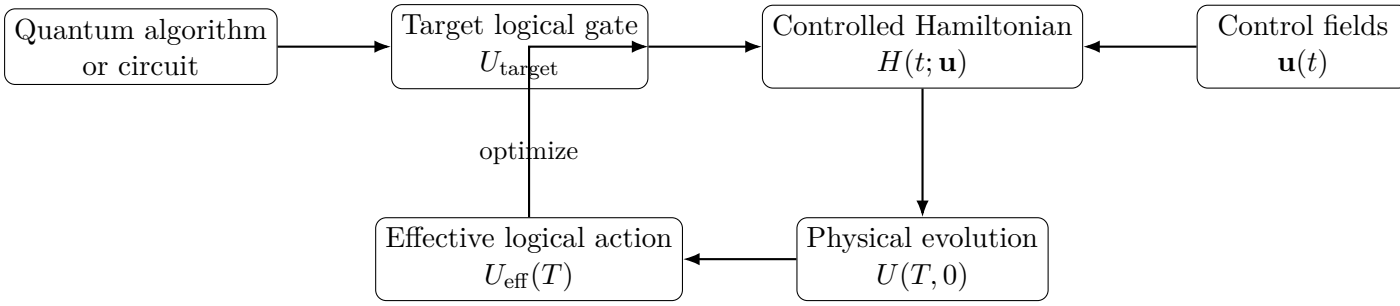

Figure 1: Conceptual relation between the algorithmic description of a quantum task and its physical realization through controlled dynamics. The central inverse problem of quantum control is to determine the control fields $\mathbf{u}(t)$ such that the induced propagator $U(T,0)$ reproduces the target logical gate on the computational subspace.

up to a physically irrelevant global phase. A convenient generic mismatch functional is therefore

$$\varepsilon_U(T) = \min_{\phi\in\mathbb{R}} \left\| U_{\text{eff}}(T) - e^{i\phi} U_{\text{target}} \right\|, \tag{9}$$

where the choice of norm depends on the application. In practical quantum-control problems, the abstract objective in Eq. (9) is usually replaced by experimentally meaningful figures of merit, such as gate fidelity, leakage penalties, smoothness constraints, and runtime penalties. These quantities will be introduced in the following paragraphs and then specialized, later in the thesis, to the UFO framework studied in this work [**?**, **?**].

For numerical optimization, it is often useful to discretize the controls on a grid $0 = t_0 < t_1 < \cdots < t_N = T$ and use a piecewise-constant parameterization,

$$u_k(t) = u_{k,\ell}, \qquad t \in [t_\ell, t_{\ell+1}), \quad k = 1,\dots,m, \quad \ell = 0,\dots,N-1. \tag{10}$$

Equation (10) turns the search for an admissible control field into a finite-dimensional optimization problem over the amplitudes $\{u_{k,\ell}\}$. This viewpoint is especially convenient because it connects naturally both with gradient-based pulse engineering and with reinforcement-learning formulations, where the control trajectory is built sequentially over discrete time steps [**?**, **?**, **?**].

0.2 Gate fidelity and runtime as central performance metrics

Once a target logical operation has been formulated as a controlled dynamical task, the next question is how to assess the quality of a candidate control protocol. In quantum control, two figures of merit play a particularly central role: the *fidelity* of the implemented gate and its *runtime*. The first quantifies how accurately the physical evolution reproduces the desired logical transformation, while the second measures how long the quantum system must remain under active control in order to realize that transformation. These quantities are not merely technical diagnostics: they directly determine the practical computational power of a hardware platform. In particular, the paper studied in this thesis explicitly identifies fidelity and runtime as the key measures governing the usefulness of a control strategy, and builds its optimization framework around their joint treatment [**?**].

Let $U_{\text{target}} \in U(d)$ denote the desired logical gate acting on a d-dimensional computational subspace, and let $U(T)$ denote the unitary evolution generated by a chosen control protocol at final time T. In the idealized closed-system setting, where the action of the control sequence remains unitary on the relevant subspace, a standard overlap-based figure of merit is the *gate fidelity*

$$F_{\text{gate}}(U(T), U_{\text{target}}) = \frac{1}{d^2} \left| \mathrm{Tr}\Big(U_{\text{target}}^\dagger U(T) \Big) \right|^2. \tag{11}$$

This quantity satisfies

$$0 \leq F_{\text{gate}} \leq 1, \tag{12}$$

and reaches its maximal value $F_{\text{gate}} = 1$ if and only if

$$U(T) = e^{i\phi} U_{\text{target}}, \qquad \phi \in \mathbb{R}, \tag{13}$$

that is, when the realized operation agrees with the target up to an irrelevant global phase. The associated *gate infidelity* is then defined as

$$\varepsilon_{\text{gate}} := 1 - F_{\text{gate}}. \tag{14}$$

This is the quantity most frequently minimized in practical control optimization, since it vanishes precisely at perfect implementation and is cheap to evaluate numerically. The uploaded paper adopts exactly this form of gate infidelity as the first contribution to its control cost function, emphasizing that it provides a computationally convenient measure of control inaccuracy [**?**].

In realistic devices, however, the implemented operation is not exactly unitary on the logical subspace, since calibration errors, stochastic fluctuations, decoherence, and leakage make the effective map better described by a quantum channel $\mathcal{E}$. In that more general setting, an experimentally meaningful notion of quality is the *average gate fidelity*, defined by averaging over all pure input states with respect to the Haar measure:

$$\overline{F}(\mathcal{E}, U_{\text{target}}) = \int d\psi \; \langle\psi| U_{\text{target}}^{\dagger} \, \mathcal{E}(|\psi\rangle\langle\psi|) \; U_{\text{target}} |\psi\rangle. \tag{15}$$

Equation (15) measures the average probability that an arbitrary input state is transformed as intended. It is therefore more operational than a purely state-specific fidelity, and it is particularly useful when one wishes to characterize a gate rather than a single state-transfer task. As discussed in the uploaded paper, average fidelity is also the natural quantity for assessing robustness under stochastic control perturbations, since one may compare how the performance of a control protocol degrades under repeated noisy realizations [**?**]. A related quantity is the *average gate infidelity*,

$$\overline{\varepsilon}_{\text{gate}} := 1 - \overline{F}(\mathcal{E}, U_{\text{target}}) \,. \tag{16}$$

For trace-preserving quantum channels, the average fidelity is closely related to the entanglement fidelity and process fidelity. In particular, if F_{e} denotes the entanglement fidelity with respect to the ideal target unitary, then one has the standard relation

$$\overline{F} = \frac{d\, F_{\text{e}} + 1}{d + 1}. \tag{17}$$

In the special case where the implemented evolution is itself unitary, Eq. (15) reduces to

$$\overline{F}(U(T), U_{\text{target}}) = \frac{\left|\text{Tr}\left(U_{\text{target}}^{\dagger} U(T)\right)\right|^2 + d}{d(d+1)}. \tag{18}$$

Hence, Eqs. (11) and (18) encode the same mismatch information up to a dimension-dependent affine transformation. This is one reason why overlap-based infidelity is so widely used in numerical optimal control: it is simpler to evaluate, while still remaining closely tied to experimentally relevant benchmarking quantities [**?**, **?**]. In the present thesis, the distinction between these notions will remain important: the optimization problem is naturally formulated in terms of gate infidelity, whereas robustness analyses are more naturally reported in terms of average fidelity under noisy implementations, exactly as done in the reference paper [**?**].

Accuracy alone, however, is not sufficient. A gate that is extremely faithful but requires too long a control time may still be of limited practical use, because quantum systems are inevitably

exposed to decoherence, relaxation, dephasing, and additional control noise throughout the entire duration of the pulse. For this reason, the total runtime T of a control protocol must be treated as a second fundamental metric. At the most basic level, runtime is simply the final integration time appearing in the propagator

$$U(T,0) = \mathcal{T}\exp\left[-\frac{i}{\hbar}\int_0^T H(t)\,dt\right], \tag{19}$$

but physically it quantifies the temporal exposure of the quantum state to all imperfections that accumulate during the gate execution. This is why the uploaded paper explicitly states that shorter gate runtimes extend the achievable circuit depth by attempting to complete the intended operation before uncorrectable errors induced by noise and dissipation become dominant [**?**].

A simple phenomenological way to express this point is to note that, in the presence of a coherence timescale T_{coh}, one generically expects the quality of a gate to deteriorate as its duration becomes a non-negligible fraction of that timescale. At the level of a rough heuristic model, one may write

$$F_{\mathrm{coh}}(T) \sim e^{-T/T_{\mathrm{coh}}}, \tag{20}$$

or, equivalently,

$$\varepsilon_{\mathrm{coh}}(T) \sim 1 - e^{-T/T_{\mathrm{coh}}}. \tag{21}$$

These expressions should not be interpreted as a complete device model, since the actual decay law depends on the physical noise channels and on the precise gate under consideration. Their role is instead conceptual: they make explicit why gate duration is not a neutral parameter. All else being equal, faster gates are generally preferable because they reduce the time window over which incoherent errors can act. This is precisely why runtime minimization is a standard objective in quantum optimal control, and why the UFO framework includes an explicit runtime penalty term in its cost function [**?**, **?**].

The role of runtime is even clearer when one considers that quantum algorithms are implemented as long sequences of elementary or composite operations. If a circuit contains N_{gates} successive gates with durations $\{T_j\}_{j=1}^{N_{\mathrm{gates}}}$, the total physical runtime is approximately

$$T_{\mathrm{circ}} = \sum_{j=1}^{N_{\mathrm{gates}}} T_j \tag{22}$$

up to idle times and synchronization overhead. Thus, improving a single gate by shortening its duration can have a multiplicative effect at the circuit level. In this sense, gate runtime is not merely a low-level engineering quantity; it is a parameter that constrains the accessible algorithmic depth before decoherence overwhelms the computation. The reference paper exploits exactly this logic when comparing learned analog controls with a discrete gate-synthesis baseline: one of its main claims is that direct pulse-level optimization can realize certain target unitaries significantly faster than their decomposition into a standard gate sequence [**?**].

The central conceptual point is therefore that quantum control is intrinsically a *multi-objective* problem. Maximizing fidelity alone may lead to excessively long or experimentally fragile pulses, while minimizing runtime alone may produce aggressive controls that suffer from leakage, control non-adiabaticity, or large sensitivity to calibration errors. In practice one seeks a compromise among several competing desiderata, with fidelity and runtime representing the two most basic ones. This tradeoff can be stated abstractly by introducing a generic control objective of the form

$$\mathcal{J}[\boldsymbol{u}] = w_F\,\varepsilon_{\mathrm{gate}}[\boldsymbol{u}] + w_T\,T[\boldsymbol{u}] + \cdots, \tag{23}$$

where $\boldsymbol{u}$ denotes the control protocol and the omitted terms may encode additional physical penalties such as leakage, bandwidth constraints, boundary conditions, or energy costs. Equation (23) is intentionally schematic, but it captures the philosophy that underlies much of modern

Table 1: Core performance metrics used in gate-level quantum control.

Quantity	Definition / interpretation
Gate fidelity	$F_{\text{gate}} = \frac{1}{d^2}\left\lvert \text{Tr}(U_{\text{target}}^\dagger U)\right\rvert^2$
Gate infidelity	$\varepsilon_{\text{gate}} = 1 - F_{\text{gate}}$
Average gate fidelity	$\overline{F}(\mathcal{E}, U_{\text{target}}) = \int d\psi\, \langle\psi\lvert U_{\text{target}}^\dagger \mathcal{E}(\lvert\psi\rangle\langle\psi\rvert) U_{\text{target}}\rvert\psi\rangle$
Average gate infidelity	$\overline{\varepsilon}_{\text{gate}} = 1 - \overline{F}$
Runtime	Final control duration T entering the propagator $U(T, 0)$

quantum control. In the present thesis, this general viewpoint will become concrete in the next chapter, where the UFO cost function will be introduced as a specific realization of this multi-objective principle, with explicit terms for infidelity, leakage, boundary constraints, and runtime [**?**].

From the viewpoint of the thesis structure, the discussion above motivates the distinction that will be used throughout the rest of the work. When the implemented evolution can be treated as approximately unitary, gate infidelity as in Eq. (14) provides a convenient optimization target. When robustness against stochastic errors is studied, average fidelity as in Eq. (15) becomes the more informative diagnostic. Runtime, on the other hand, remains a fundamental metric in both settings, because it directly mediates the tradeoff between fast control and exposure to imperfections. Together, these quantities provide the minimal language needed to formulate the control problem addressed in this thesis.

0.3 Universal gates and direct analog control

A central notion in quantum computation is that of *universality*. At the abstract circuit level, one usually says that a collection of elementary gates $\mathcal{G}$ is universal if arbitrary unitary operations on a finite number of qubits can be generated—exactly or to arbitrary accuracy—by composing gates from $\mathcal{G}$ [**?**, **?**]. For an n-qubit register with logical Hilbert space

$$\mathcal{H}_{\text{comp}} \cong (\mathbb{C}^2)^{\otimes n}, \tag{24}$$

the ideal set of logical operations is the unitary group $U(2^n)$, or equivalently $SU(2^n)$ up to a global phase. A finite gate set $\mathcal{G}$ is therefore called universal when the subgroup it generates is dense in $SU(2^n)$:

$$\overline{\langle \mathcal{G} \rangle} = SU(2^n). \tag{25}$$

Equation (25) formalizes the standard circuit-model view: even if the hardware natively exposes only a limited menu of primitive operations, it is still computationally sufficient provided those primitives can approximate any desired logical transformation to arbitrary precision.

In this framework, a target gate $U_{\text{target}} \in SU(2^n)$ is not implemented directly, but rather compiled into a finite sequence of elementary operations,

$$U_{\text{syn}} = G_L G_{L-1} \cdots G_1, \qquad G_\ell \in \mathcal{G}, \tag{26}$$

such that

$$\left\| U_{\text{target}} - e^{i\phi} U_{\text{syn}} \right\| \leq \varepsilon, \qquad \phi \in \mathbb{R}, \tag{27}$$

for a prescribed synthesis accuracy $\varepsilon > 0$. The existence of such approximations for universal sets is guaranteed in general, and the Solovay–Kitaev theorem further shows that efficient approximations can be constructed with sequence length growing only polylogarithmically in $1/\varepsilon$ for standard assumptions on the gate set [**?**, **?**]. From the perspective of fault-tolerant quantum

computing, this discrete picture is fundamental: logical algorithms are compiled into a standard finite library of protected primitives, and the abstraction is both mathematically elegant and operationally robust.

However, the circuit-synthesis viewpoint is not the only physically relevant one. In an actual device, each elementary gate G_ℓ must itself be realized by controlled time evolution. Thus, the synthesized circuit in Eq. (26) ultimately corresponds to a concatenation of pulse segments, each with its own control fields, calibration overhead, and execution time. If τ_ℓ denotes the runtime of the ℓ-th elementary gate, the total physical duration of the synthesized implementation is approximately

$$T_{\mathrm{syn}} = \sum_{\ell=1}^{L} \tau_\ell, \tag{28}$$

while the logical depth is the number of time layers required to schedule these operations,

$$D_{\mathrm{syn}} \leq L. \tag{29}$$

Equation (28) makes explicit a simple but important fact: even when the number of compiled gates is information-theoretically optimal, the resulting implementation may still be physically expensive, because every additional gate consumes coherence time and introduces further opportunities for control error and leakage. This observation becomes especially relevant in pre-fault-tolerant and noisy intermediate-scale quantum devices, where the experimentally accessible circuit depth is strongly limited by decoherence and imperfect control [**?**].

For this reason, near-term hardware often benefits from a more direct viewpoint, in which one seeks to realize the full target unitary through a single analog control protocol rather than through a discrete decomposition. In the notation introduced earlier, one instead searches for control fields $\boldsymbol{u}(t)$ such that

$$U(T,0) = \mathcal{T}\exp\left[-\frac{i}{\hbar}\int_0^T H(t;\boldsymbol{u})\, dt\right] \approx e^{i\phi} U_{\mathrm{target}}, \tag{30}$$

with T as small as possible and with all relevant physical constraints satisfied. The contrast with Eq. (26) is conceptual as much as technical. In circuit synthesis, the target unitary is approximated through a universal symbolic alphabet of gates. In direct analog control, by contrast, one exploits the continuous dynamics naturally generated by the underlying Hamiltonian in order to realize the target in one coherent control episode. The uploaded paper is explicitly motivated by this distinction: for near-term devices, it argues that quantum operations need not be restricted to a finite universal gate set, and that implementing a desired unitary with one pulse sequence can reduce resource overhead relative to a longer synthesized circuit [**?**].

This distinction is closely tied to the notion of *controllability*. Let the controlled Hamiltonian be written as

$$H(t) = H_{\mathrm{d}} + \sum_{k=1}^{m} u_k(t) H_k, \tag{31}$$

where H_{d} is the drift term and the H_k are control generators. A standard sufficient condition for full operator controllability on a d-dimensional Hilbert space is that the dynamical Lie algebra generated by the skew-Hermitian operators $\{iH_{\mathrm{d}}, iH_1, \ldots, iH_m\}$ spans the whole algebra $\mathfrak{su}(d)$:

$$\mathrm{Lie}\{iH_{\mathrm{d}}, iH_1, \ldots, iH_m\} = \mathfrak{su}(d). \tag{32}$$

Equation (32) expresses the ideal mathematical limit of analog universality: if the available drift and control Hamiltonians generate all of $\mathfrak{su}(d)$, then any unitary in $SU(d)$ is reachable in principle through suitable time-dependent controls [**?**]. In practice, of course, reachability alone is not enough. One must still contend with finite amplitudes, bandwidth limits, leakage outside the computational subspace, calibration uncertainty, and noise. These constraints make the

effective implementation of arbitrary target gates highly nontrivial even when controllability holds in the abstract.

The practical advantage of direct analog control is most pronounced when the target unitary is naturally aligned with the native interactions of the hardware. If a device Hamiltonian contains, for example, an entangling term of the form

$$H_{\text{nat}} = J\left(\sigma_1^x\sigma_2^x + \sigma_1^y\sigma_2^y\right), \tag{33}$$

then free or weakly modulated evolution under this interaction generates the gate family

$$U_{\text{nat}}(t) = \exp\left[-\frac{i}{\hbar}tH_{\text{nat}}\right] = \exp[-i\theta(t)\left(\sigma_1^x\sigma_2^x + \sigma_1^y\sigma_2^y\right)], \tag{34}$$

with $\theta(t) = Jt/\hbar$. In such a situation, certain entangling operations may be realized almost directly by the device physics itself, possibly supplemented by a small number of local rotations, instead of being compiled into a longer generic circuit. This is precisely the type of hardware-structure exploitation that makes analog control attractive: the control procedure can adapt to the geometry and couplings of a given platform rather than forcing every task through a fixed universal template [**?**, **?**].

From a computational perspective, one may therefore distinguish two different notions of efficiency. The first is *gate-set efficiency*, meaning how economically a target unitary can be decomposed into a prescribed universal library. The second is *hardware efficiency*, meaning how economically that same unitary can be realized on a specific physical platform once its native interactions are taken into account. These notions need not coincide. A gate may admit an optimal decomposition within a discrete universal set and yet still be slower than a direct analog realization that exploits the natural Hamiltonian of the device. This tension is one of the central motivations of the present thesis, and it is made explicit in the uploaded paper, which compares direct analog controls with optimal gate-synthesis counterparts for a family of two-qubit gates and reports significant runtime reductions for some targets [**?**]. In the following chapters, this distinction will become concrete in the superconducting-gmon setting, where a family of physically relevant two-qubit gates can be analyzed both from the perspective of optimal discrete synthesis and from that of direct pulse-level realization.

It is important, however, not to overstate the case. Direct analog control does not replace the circuit model as a fundamental description of quantum computation, nor does it eliminate the need for universal gate sets in fault-tolerant architectures. Rather, it provides a complementary layer of optimization that is particularly valuable in near-term devices and in hardware-aware quantum algorithms. The relevant question is not whether one should abandon universal discrete gates, but whether a given target operation should be implemented through a compiled sequence or through a direct continuous control trajectory. The answer depends on the interaction between algorithmic structure, hardware native couplings, noise, and control constraints. This is exactly the regime in which modern quantum-control methods, including reinforcement-learning-based approaches, become especially relevant [**?**, **?**].

0.4 Superconducting qubits as weakly anharmonic multilevel systems

Among the leading physical platforms for gate-based quantum information processing, superconducting circuits occupy a particularly important position because they combine lithographic scalability, strong microwave control, and compatibility with fast entangling interactions [**?**, **?**]. From the theoretical point of view, however, a superconducting qubit is not fundamentally a two-level system. Rather, it is a quantized nonlinear electrical circuit built from capacitive, inductive, and Josephson elements. The Josephson junction provides the essential nonlinearity that breaks the equal spacing of a harmonic oscillator and thereby allows the two lowest energy eigenstates to be selectively addressed as a logical qubit [**?**, **?**]. This observation is central for the

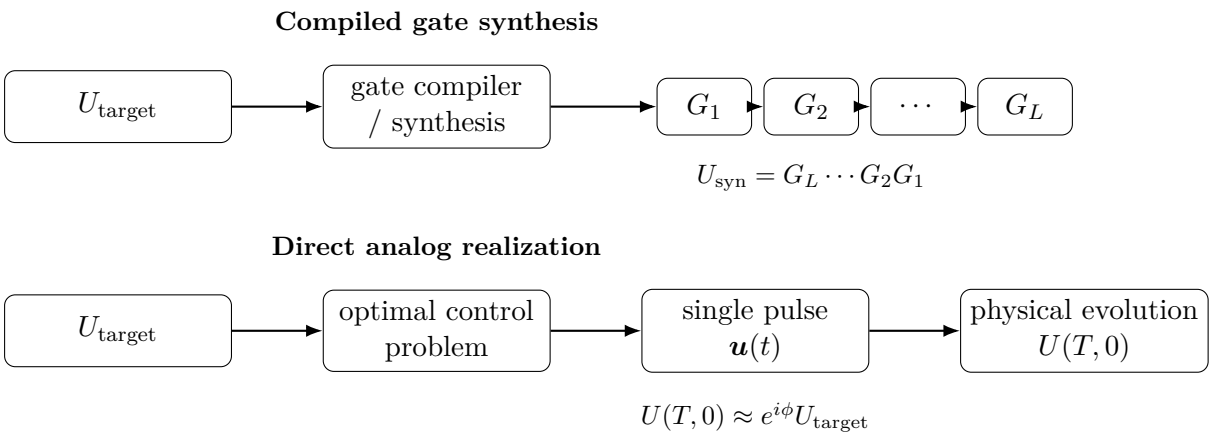

Figure 2: Two conceptually distinct routes to implementing a target unitary. In the upper route, the gate is compiled into a sequence of elements from a finite universal set. In the lower route, one directly optimizes a continuous control protocol so that the full physical evolution approximates the target in a single coherent pulse sequence.

present thesis, because the control problem studied later is formulated on a hardware platform of this kind, and the possibility of populating levels outside the logical subspace is precisely what gives rise to leakage errors.

A standard starting point is the circuit Hamiltonian of a Cooper-pair-box or transmon-type qubit,

$$\hat{H}_{\text{tr}} = 4E_C(\hat{n} - n_g)^2 - E_J \cos\hat{\varphi}, \tag{35}$$

where E_C is the charging energy, E_J the Josephson energy, $\hat{n}$ the number operator for transferred Cooper pairs, $\hat{\varphi}$ the superconducting phase difference across the junction, and n_g an offset charge [**?**]. The canonical variables satisfy

$$[\hat{\varphi}, \hat{n}] = i, \tag{36}$$

so that Eq. (35) describes a nonlinear quantum oscillator. In the transmon regime,

$$\frac{E_J}{E_C} \gg 1, \tag{37}$$

charge dispersion is exponentially suppressed, which strongly reduces sensitivity to charge noise, while the residual nonlinearity remains sufficient for qubit operation [**?**]. This is one of the main reasons why transmon-like devices, including xmon- and gmon-based architectures, have become so prominent in superconducting quantum computing [**?**, **?**, **?**].

The physics of weak anharmonicity becomes transparent by expanding the cosine potential around one of its minima,

$$-E_J \cos\hat{\varphi} = -E_J + \frac{E_J}{2}\hat{\varphi}^2 - \frac{E_J}{24}\hat{\varphi}^4 + O(\hat{\varphi}^6). \tag{38}$$

Substituting Eq. (38) into Eq. (35), one obtains to lowest nontrivial order

$$\hat{H}_{\text{tr}} \approx 4E_C\hat{n}^2 + \frac{E_J}{2}\hat{\varphi}^2 - \frac{E_J}{24}\hat{\varphi}^4 - E_J. \tag{39}$$

The first two terms define a harmonic oscillator, while the quartic correction introduces a weak nonlinearity. It is therefore natural to interpret the device as an oscillator whose level spacing is almost, but not exactly, uniform. In other words, superconducting transmon-like qubits are *weakly anharmonic oscillators* rather than exact spin-1/2 systems [**?**, **?**]. This distinction is

crucial: without anharmonicity, a resonant drive tuned to the $|0\rangle \leftrightarrow |1\rangle$ transition would also excite higher transitions, making selective qubit control impossible.

Introducing bosonic ladder operators in the usual way,

$$\hat{\varphi} = \varphi_{\text{zpf}}(\hat{a} + \hat{a}^\dagger), \qquad \hat{n} = i n_{\text{zpf}}(\hat{a}^\dagger - \hat{a}), \tag{40}$$

with zero-point fluctuation amplitudes chosen so as to diagonalize the quadratic part, one may rewrite the Hamiltonian approximately as a weakly nonlinear oscillator,

$$\hat{H}_{\text{tr}} \approx \omega_p \left(\hat{a}^\dagger \hat{a} + \frac{1}{2}\right) + \frac{\alpha}{2} \hat{a}^\dagger \hat{a}^\dagger \hat{a}\hat{a} + \cdots, \tag{41}$$

where

$$\omega_p \approx \sqrt{8E_J E_C}, \tag{42}$$

in units with $\hbar = 1$, and where α denotes the anharmonicity. More precisely, perturbation theory yields the well-known large-E_J/E_C approximation for the energy levels

$$E_m \approx -E_J + \sqrt{8E_J E_C}\left(m + \frac{1}{2}\right) - \frac{E_C}{12}\left(6m^2 + 6m + 3\right), \qquad m = 0, 1, 2, \ldots, \tag{43}$$

from which the transition frequencies follow as

$$\omega_{m,m+1} = E_{m+1} - E_m \approx \sqrt{8E_J E_C} - E_C(m+1). \tag{44}$$

In particular,

$$\omega_{01} \approx \sqrt{8E_J E_C} - E_C, \qquad \omega_{12} \approx \sqrt{8E_J E_C} - 2E_C, \tag{45}$$

so that the anharmonicity is approximately

$$\alpha := \omega_{12} - \omega_{01} \approx -E_C. \tag{46}$$

Equation (46) shows explicitly why the transmon is only weakly anharmonic: the nonuniformity of the spectrum is governed by E_C, while the overall transition frequency is of order $\sqrt{8E_J E_C}$. In the transmon regime of Eq. (37), one therefore has

$$|\alpha| \ll \omega_{01}, \tag{47}$$

which is the mathematical expression of the weakly anharmonic limit [**?**, **?**].

The logical qubit is then encoded in the two-dimensional computational subspace

$$\mathcal{H}_{\text{comp}} = \text{span}\left\{|0\rangle, |1\rangle\right\}, \tag{48}$$

where $|0\rangle$ and $|1\rangle$ denote the ground and first excited eigenstates of the circuit Hamiltonian. The full device Hilbert space, however, is larger:

$$\mathcal{H}_{\text{phys}} = \text{span}\left\{|0\rangle, |1\rangle, |2\rangle, |3\rangle, \ldots\right\}. \tag{49}$$

Thus, the qubit approximation is not fundamental but effective: it consists in restricting the dynamics to Eq. (48) and neglecting population transfer into higher excited states. This approximation is excellent when control pulses are sufficiently weak, spectrally selective, and adiabatic with respect to unwanted transitions, but it is never exact. In particular, short or strong pulses can admix the states $|2\rangle, |3\rangle, \ldots$, making the multilevel structure physically relevant even for nominally two-level gate operations [**?**, **?**].

For this reason, it is often useful to describe a driven superconducting qubit directly in bosonic language. A generic driven weakly anharmonic mode may be modeled as

$$\hat{H}(t) = \omega_q \hat{a}^\dagger \hat{a} + \frac{\alpha}{2} \hat{a}^\dagger \hat{a}^\dagger \hat{a}\hat{a} + \Omega(t)\left(\hat{a} + \hat{a}^\dagger\right), \tag{50}$$

where ω_q is the fundamental transition frequency, $\alpha < 0$ the weak anharmonicity, and $\Omega(t)$ an externally applied microwave drive. Equation (50) makes two points transparent. First, the creation and annihilation operators act on the full ladder of states and therefore naturally include higher-level dynamics. Second, the same control field that is intended to drive the $|0\rangle \leftrightarrow |1\rangle$ transition also couples, in principle, to $|1\rangle \leftrightarrow |2\rangle$ and higher transitions, with selectivity arising only because the spectrum is not exactly harmonic. The control task is therefore intrinsically multilevel.

This multilevel perspective is not merely a pedagogical refinement; it is the viewpoint adopted in the reference paper analyzed in this thesis. In its Appendix A, the gmon architecture is modeled by quantizing the circuit degrees of freedom in terms of bosonic modes and then deriving an effective rotating-wave Hamiltonian with explicit anharmonicity, tunable coupling, detuning controls, and microwave drives [**?**]. In that formulation, each superconducting qubit is treated as a weakly anharmonic oscillator before projection onto the qubit basis, exactly in line with the general discussion above. This is the correct framework for studying high-performance control, because it preserves the higher excited levels that are responsible for leakage and because it allows one to assess whether a pulse remains confined to the intended logical sector.

When two such devices are coupled, the same principle persists at the multi-qubit level. If each qubit is truncated to d_{loc} levels, the full two-body Hilbert space has dimension

$$\dim \mathcal{H}^{(2)}_{\text{phys}} = d_{\text{loc}}^2, \tag{51}$$

whereas the logical computational sector retains dimension

$$\dim \mathcal{H}^{(2)}_{\text{comp}} = 4. \tag{52}$$

Already with $d_{\text{loc}} = 3$, one obtains a 9-dimensional physical space for a nominally two-qubit gate. This illustrates why the multilevel description rapidly becomes unavoidable in realistic control simulations: even if the target gate acts only on the logical subspace, the actual Schrödinger evolution takes place in a larger Hilbert space that contains leakage channels.

In summary, superconducting qubits should be viewed as engineered nonlinear oscillators whose weak anharmonicity enables qubit selectivity but does not eliminate the presence of higher excited states. The logical qubit is encoded in the two lowest levels, yet the full control problem is inherently multilevel. This fact underlies the entire structure of modern superconducting quantum control: it explains why one uses bosonic or multilevel Hamiltonians at the modeling stage, why fast control can induce unwanted excitations, and why leakage must later be treated as an explicit optimization objective. The next paragraph will build directly on this point by introducing the computational subspace more carefully and defining leakage as population transfer outside it.

0.5 Computational subspace and leakage

The effective qubit description of a superconducting circuit is obtained by selecting a low-dimensional sector of the full Hilbert space and declaring it to be the logical space used for computation. For a single weakly anharmonic mode, this logical sector is the span of the two lowest eigenstates,

$$\mathcal{H}^{(1)}_{\text{comp}} = \text{span}\{|0\rangle, |1\rangle\}, \tag{53}$$

while the full physical Hilbert space remains

$$\mathcal{H}^{(1)}_{\text{phys}} = \text{span}\{|0\rangle, |1\rangle, |2\rangle, |3\rangle, \dots\}. \tag{54}$$

For a register of n such devices, the logical computational subspace is the tensor product of the local qubit sectors,

$$\mathcal{H}_{\text{comp}} = \left(\mathcal{H}^{(1)}_{\text{comp}}\right)^{\otimes n} \cong (\mathbb{C}^2)^{\otimes n}, \tag{55}$$

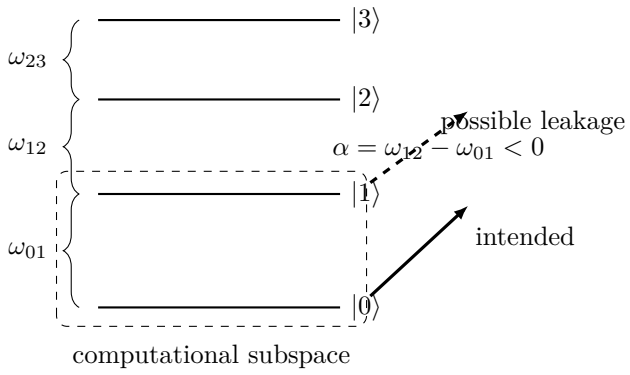

Figure 3: Schematic spectrum of a weakly anharmonic superconducting qubit. The two lowest levels define the computational subspace, while the higher levels remain physically present. Because the transition spacings are unequal but not widely separated, a drive resonant with the logical transition can still weakly couple to higher transitions, especially under strong or short pulses.

with dimension

$$d_{\text{comp}} = 2^n. \tag{56}$$

The full physical space is larger,

$$\mathcal{H}_{\text{phys}} = \mathcal{H}_{\text{comp}} \oplus \mathcal{H}_{\text{leak}}, \tag{57}$$

where $\mathcal{H}_{\text{leak}}$ collects all states outside the intended qubit sector. The decomposition in Eq. (57) is conceptually central: a quantum gate is specified only on $\mathcal{H}_{\text{comp}}$, but the actual time evolution takes place on $\mathcal{H}_{\text{phys}}$ [**?**, **?**, **?**].

It is therefore natural to introduce the orthogonal projectors onto the computational and leakage sectors,

$$\Pi_{\text{comp}} = \sum_{\mu=1}^{d_{\text{comp}}} |\mu\rangle \langle\mu| , \qquad \Pi_{\text{leak}} = \mathbb{I} - \Pi_{\text{comp}}, \tag{58}$$

where $\{|\mu\rangle\}$ is any orthonormal basis of $\mathcal{H}_{\text{comp}}$. For a state $|\psi(t)\rangle \in \mathcal{H}_{\text{phys}}$, initialized inside the computational subspace,

$$\Pi_{\text{comp}} |\psi(0)\rangle = |\psi(0)\rangle , \tag{59}$$

the instantaneous leakage probability is defined as

$$L_\psi(t) := \langle\psi(t)|\Pi_{\text{leak}}|\psi(t)\rangle = 1 - \langle\psi(t)|\Pi_{\text{comp}}|\psi(t)\rangle. \tag{60}$$

Equation (60) gives the most direct meaning of leakage: it is simply the population that has escaped the logical sector. Since Π_{leak} is a projector, one has

$$0 \le L_\psi(t) \le 1. \tag{61}$$

If $L_\psi(t) = 0$, the state remains fully inside $\mathcal{H}_{\text{comp}}$; if $L_\psi(t) > 0$, then part of the wavefunction has reached states that are outside the intended qubit encoding.

At the gate level one is interested not in a single initial state, but in how well the full control protocol preserves the computational subspace on average. Let $U(t, 0)$ denote the physical unitary evolution generated by the driven multilevel Hamiltonian. For an arbitrary pure input state $|\psi\rangle \in \mathcal{H}_{\text{comp}}$, the survival probability in the computational subspace after time t is

$$P_{\text{stay}}(\psi, t) = \langle\psi|U^\dagger(t, 0)\, \Pi_{\text{comp}}\, U(t, 0)|\psi\rangle, \tag{62}$$

and the corresponding leakage is

$$L(\psi, t) = 1 - P_{\text{stay}}(\psi, t). \tag{63}$$

A natural averaged quantity is obtained by integrating over all pure states in the computational subspace with respect to the Haar measure,

$$\overline{L}(t) = 1 - \int_{\mathcal{H}_{\text{comp}}} d\psi \, \langle\psi|U^\dagger(t,0)\,\Pi_{\text{comp}}\,U(t,0)|\psi\rangle. \tag{64}$$

Using the identity

$$\int_{\mathcal{H}_{\text{comp}}} d\psi \, \langle\psi|A|\psi\rangle = \frac{1}{d_{\text{comp}}}\text{Tr}_{\text{comp}}(A), \tag{65}$$

valid for any operator A acting on $\mathcal{H}_{\text{comp}}$, Eq. (64) becomes

$$\overline{L}(t) = 1 - \frac{1}{d_{\text{comp}}}\text{Tr}\Big(\Pi_{\text{comp}}U^\dagger(t,0)\Pi_{\text{comp}}U(t,0)\Pi_{\text{comp}}\Big). \tag{66}$$

Equation (66) is especially useful in numerical simulations, because it directly quantifies the average fraction of logical population that fails to remain in the computational sector.

To understand the origin of leakage, it is useful to decompose the full Hamiltonian with respect to the direct sum in Eq. (57). Writing

$$H(t) = \Pi_{\text{comp}}H(t)\Pi_{\text{comp}} + \Pi_{\text{leak}}H(t)\Pi_{\text{leak}} + \Pi_{\text{comp}}H(t)\Pi_{\text{leak}} + \Pi_{\text{leak}}H(t)\Pi_{\text{comp}}, \tag{67}$$

one may define the block-diagonal part

$$H_{\text{diag}}(t) := \Pi_{\text{comp}}H(t)\Pi_{\text{comp}} + \Pi_{\text{leak}}H(t)\Pi_{\text{leak}}, \tag{68}$$

and the block-off-diagonal part

$$H_{\text{off}}(t) := \Pi_{\text{comp}}H(t)\Pi_{\text{leak}} + \Pi_{\text{leak}}H(t)\Pi_{\text{comp}}. \tag{69}$$

The interpretation is immediate: $H_{\text{diag}}(t)$ generates dynamics *within* each sector, whereas $H_{\text{off}}(t)$ couples the computational subspace to the leakage sector and is therefore the fundamental microscopic source of leakage. In this sense, leakage is not an external add-on to the control problem; it is already encoded in the multilevel structure of the Hamiltonian itself [**?**, **?**, **?**].

This block structure is particularly transparent in the two-qubit gmon model studied later in this thesis. There, the computational subspace is

$$\Omega_0 = \text{span}\{|00\rangle, |10\rangle, |01\rangle, |11\rangle\}, \tag{70}$$

while higher-excitation sectors, such as

$$\Omega_1 = \text{span}\{|20\rangle, |21\rangle, |12\rangle, |02\rangle\}, \tag{71}$$

belong to the leakage space. In the notation of the reference paper, the block-off-diagonal Hamiltonian term is precisely the component that couples different energy subspaces and is identified as the culprit behind leakage errors [**?**]. The same paper further emphasizes that the relevant energy-scale separation is between intra-subspace couplings and the gap Δ between Ω_0 and higher sectors, which is why leakage becomes especially severe when controls are too strong or too rapidly modulated [**?**].

A further distinction, important both conceptually and practically, is that between *coherent* and *incoherent* leakage. Coherent leakage refers to a reversible unitary transfer of amplitude between $\mathcal{H}_{\text{comp}}$ and $\mathcal{H}_{\text{leak}}$, generated by the Hamiltonian couplings in Eq. (69). Since the full evolution remains unitary, the leaked amplitude can in principle return to the computational

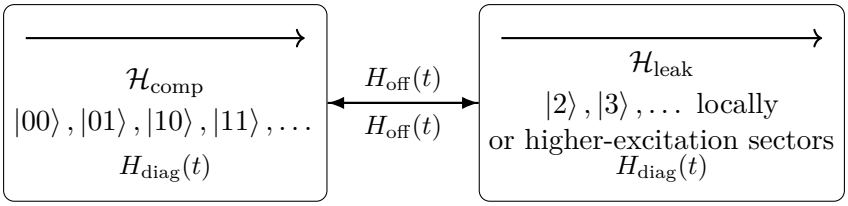

Figure 4: Schematic decomposition of the physical Hilbert space into computational and leakage sectors. The block-diagonal Hamiltonian generates evolution within each sector, while the block-off-diagonal part $H_{\text{off}}(t)$ couples the two sectors and is the microscopic source of leakage.

sector at a later time. In contrast, incoherent leakage denotes effectively irreversible loss from the computational subspace, for example through decay to environmental modes, photon loss, or other open-system processes. In the language of the uploaded paper, even non-adiabatic excursions induced during time-dependent modulation may be treated as effectively incoherent in practice when the backflow to the computational subspace does not occur on the relevant control timescale [**?**].

This distinction matters because coherent and incoherent leakage affect gate design differently. Coherent leakage may partially cancel through interference, and therefore depends sensitively on pulse shape, phase accumulation, and gate duration. Incoherent leakage, by contrast, accumulates more like a genuine loss mechanism and directly degrades the logical channel implemented on the qubit subspace. A control pulse can therefore have high apparent fidelity on a restricted logical model and yet still perform poorly on the real device if it transiently or permanently populates higher levels. This is one of the main reasons why modern pulse-engineering methods for superconducting qubits explicitly account for leakage rather than optimizing only a projected two-level model [**?**, **?**, **?**].

It is also useful to distinguish leakage from ordinary logical errors that remain inside the computational subspace. If a pulse maps $|0\rangle$ and $|1\rangle$ to the wrong superpositions of $|0\rangle$ and $|1\rangle$, then the error is purely logical and can still be described by an incorrect unitary acting on $\mathcal{H}_{\text{comp}}$. Leakage is different: once population reaches $\mathcal{H}_{\text{leak}}$, the resulting process is no longer representable as a unitary gate on the logical Hilbert space alone. This means that gate infidelity and leakage quantify genuinely different failure modes, and both must be monitored separately in any realistic multilevel control problem.

For this reason, the preservation of the computational subspace should be regarded as a structural requirement of quantum control, not merely as a secondary diagnostic. The goal is not only to approximate a target logical unitary but to do so while keeping the evolution effectively confined to $\mathcal{H}_{\text{comp}}$. In the remainder of the thesis, this idea will reappear in a more quantitative form through explicit leakage penalties and bounds. In particular, the UFO framework studied later augments the gate-infidelity objective by a dedicated leakage term constructed from the block-off-diagonal dynamics of the full multilevel Hamiltonian [**?**].

0.6 The speed–leakage tradeoff

A central difficulty of quantum control in weakly anharmonic superconducting devices is that the desire for short gate times competes directly with the need to suppress leakage outside the computational subspace. In principle, faster gates are desirable because they reduce the exposure of the system to decoherence, relaxation, dephasing, and low-frequency calibration drift. In practice, however, the implementation of a shorter gate typically requires either stronger control amplitudes, more abrupt temporal modulation, or both. In multilevel systems, these same features tend to increase the coupling to unwanted transitions and therefore raise the probability of populating non-computational states [**?**, **?**, **?**]. This tension is one of the defining constraints of pulse engineering in superconducting qubits and is a key reason why realistic

control optimization must treat runtime and leakage simultaneously rather than as independent objectives [?].

The origin of this tradeoff can already be seen in the simplest driven weakly anharmonic model. Consider a three-level truncation of a superconducting mode, with logical states $|0\rangle$ and $|1\rangle$ and the first leakage level $|2\rangle$. In a frame rotating resonantly with the $|0\rangle \leftrightarrow |1\rangle$ transition, a standard effective Hamiltonian takes the form

$$\hat{H}_{3\mathrm{lvl}}(t) = \frac{\Omega(t)}{2}\left(|0\rangle\langle 1| + |1\rangle\langle 0| + \lambda\,|1\rangle\langle 2| + \lambda\,|2\rangle\langle 1|\right) + \alpha\,|2\rangle\langle 2|\,, \tag{72}$$

where $\Omega(t)$ is the drive envelope, $\alpha = \omega_{12} - \omega_{01} < 0$ is the anharmonicity, and λ is the relative matrix element for the $|1\rangle \leftrightarrow |2\rangle$ transition. Equation (72) makes the problem explicit: the same drive that implements logical rotations also couples to the leakage transition, which is only detuned by the finite anharmonicity α. Thus, unlike in an ideal two-level system, faster control does not merely accelerate the intended dynamics; it also enhances the unwanted coupling to higher levels [?, ?, ?].

In the perturbative off-resonant regime,

$$|\Omega(t)| \ll |\alpha|, \tag{73}$$

the unwanted excitation of the leakage level can be estimated heuristically from standard off-resonant driving arguments. For a drive amplitude Ω that is approximately constant over the relevant interval, the leakage amplitude scales as

$$|c_2(t)| \sim \frac{\lambda\,\Omega}{2|\alpha|}, \tag{74}$$

so that the corresponding leakage probability obeys the rough scaling law

$$P_{\mathrm{leak}} \sim \left(\frac{\lambda\,\Omega}{2|\alpha|}\right)^2. \tag{75}$$

Although Eqs. (74)–(75) are only heuristic estimates, they capture an essential fact: increasing the control amplitude in order to shorten the gate time tends to increase leakage quadratically relative to the small parameter $\Omega/|\alpha|$. This is precisely why the short-time regime is especially challenging in weakly nonlinear oscillators, and why the leading error mechanism for fast gates is often leakage rather than an ordinary logical over-rotation [?, ?].

A complementary perspective comes from the time-frequency structure of the pulse. A control signal with characteristic duration T_{g} necessarily has a spectral width of order

$$\Delta\omega_{\mathrm{pulse}} \sim \frac{1}{T_{\mathrm{g}}}, \tag{76}$$

so that shorter pulses possess broader frequency content. Spectral selectivity of the logical transition requires that the pulse bandwidth remain small compared with the detuning to the nearest unwanted transition. In weakly anharmonic qubits, this means roughly

$$\Delta\omega_{\mathrm{pulse}} \ll |\alpha|, \tag{77}$$

or equivalently,

$$T_{\mathrm{g}} \gg \frac{1}{|\alpha|}. \tag{78}$$

Equation (78) should not be interpreted as a rigorous lower bound on the gate time, since shaped pulses and interference effects can do substantially better than naive square-pulse estimates. Its conceptual role is instead to show that fast gating is constrained by spectral crowding: once the

pulse becomes too short, its Fourier components inevitably overlap the leakage transition and the two-level approximation begins to break down [**?**, **?**].

The same conclusion can be reached from an adiabaticity viewpoint. As discussed in the previous subsection, the full Hamiltonian may be decomposed into block-diagonal and block-off-diagonal components with respect to the splitting $\mathcal{H}_{\text{phys}} = \mathcal{H}_{\text{comp}} \oplus \mathcal{H}_{\text{leak}}$. The block-off-diagonal term $H_{\text{off}}(t)$ couples the computational and leakage sectors, while rapid temporal modulation enhances non-adiabatic transfer between them. In the uploaded paper, this idea is made quantitative through a leakage bound derived within a time-dependent Schrieffer–Wolff framework, where the accumulated coherent leakage is controlled not only by the size of the block-off-diagonal Hamiltonian but also by the rate at which it varies in time. In schematic form, their bound reads

$$L_{\text{tot}} \lesssim \frac{\|\hat{H}_{\text{od}}(0)\|}{\Delta(0)} + \frac{\|\hat{H}_{\text{od}}(T)\|}{\Delta(T)} + \int_0^T \frac{1}{\Delta(t)^2} \left\| \frac{d^2 \hat{H}_{\text{od}}(t)}{dt^2} \right\| dt, \tag{79}$$

where $\Delta(t)$ denotes the relevant energy gap and $\hat{H}_{\text{od}}(t)$ the effective block-off-diagonal Hamiltonian in the transformed basis [**?**]. Equation (79) is highly instructive for the present discussion: it shows that leakage is aggravated both by large off-diagonal couplings and by rapid pulse modulation, since faster pulses typically make the derivative term larger. Thus, trying to shorten the gate time too aggressively can increase leakage even if the control amplitudes themselves remain moderate.

The speed–leakage tradeoff is therefore not merely a matter of stronger fields; it is also a matter of pulse smoothness and spectral engineering. This is the physical motivation behind derivative-based correction schemes such as DRAG, where an additional quadrature proportional to the time derivative of the main pulse is introduced to cancel leakage-inducing terms order by order [**?**, **?**]. More generally, modern optimal-control methods shape the pulse so that fast logical evolution is achieved while destructive interference suppresses net population transfer to higher levels. Experimental work on superconducting transmons confirms that this balance is nontrivial: very short gates can indeed be realized, but only with carefully optimized pulse shapes that explicitly control leakage [**?**].

At the level of control design, the existence of this tradeoff implies that gate optimization is intrinsically multi-objective. Let $\boldsymbol{u}(t)$ denote the control protocol and $T[\boldsymbol{u}]$ its final runtime. Then an abstract cost functional consistent with the preceding discussion is

$$\mathcal{J}[\boldsymbol{u}] = w_F\, \varepsilon_{\text{gate}}[\boldsymbol{u}] + w_L\, L[\boldsymbol{u}] + w_T\, T[\boldsymbol{u}] + \cdots, \tag{80}$$

where $\varepsilon_{\text{gate}}$ measures logical inaccuracy, $L[\boldsymbol{u}]$ is a leakage functional, and the omitted terms may encode boundary constraints, bandwidth penalties, fluence costs, or robustness objectives. Equation (80) summarizes the central message of this subsection: making a gate faster is not automatically beneficial unless the resulting increase in leakage remains under control. This perspective is directly aligned with the philosophy of the UFO framework, which later combines gate infidelity, leakage, control constraints, and total runtime into a single optimization objective [**?**].

From the viewpoint of superconducting-qubit physics, one may summarize the tradeoff as follows. Long pulses are usually easier to make spectrally selective and therefore tend to suppress leakage, but they suffer from greater exposure to decoherence and slow noise. Very short pulses reduce the total interaction time with the environment, but they broaden the pulse spectrum, enhance non-adiabaticity, and increase the risk of population transfer into higher levels. The practically relevant regime is therefore neither the slowest nor the fastest one in an absolute sense, but the regime in which runtime has been reduced as much as possible while leakage and control-induced errors remain acceptably small. This balance is one of the central organizing principles of high-fidelity gate design in weakly anharmonic superconducting systems [**?**, **?**, **?**].

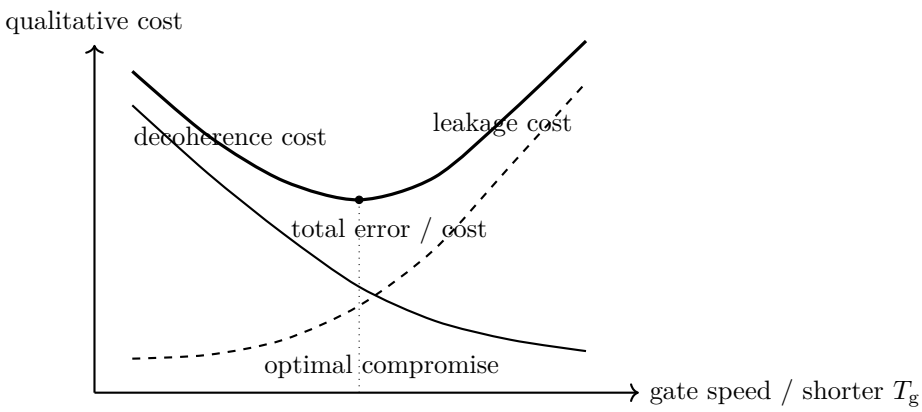

Figure 5: Qualitative illustration of the speed–leakage tradeoff in weakly anharmonic superconducting qubits. Slower gates suffer more from decoherence and low-frequency noise, whereas faster gates typically increase leakage through stronger driving, broader spectral content, and larger non-adiabatic effects. The physically relevant control problem is therefore a compromise between these competing tendencies.